

Advanced topics in seasonal adjustment: Revision policies

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■ Agenda

■ Motivation

■ Sources of revisions

■ Revision policies

■ JDemetra+ and CompRes

■ Summary

■ Useful links

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I Motivation

New data: updates and revisions

Implications for seasonal adjustment

Question: How to deal with it?

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Sources of revisions I

Data-driven changes

- Estimates, imputed figures $\rightsquigarrow$ substitution
- Errors $\rightsquigarrow$ corrections
- Annual information $\rightsquigarrow$ adjustments

Methodological changes

- Definitions, classifications
- Recording/estimation methods $\rightsquigarrow$ refinement, harmonisation
- Indices $\rightsquigarrow$ base year switch (with new weighting scheme)

■ Sources of revisions II

Availability

- New Data

Technical Changes

- Outliers, calendar effects $\rightsquigarrow$ treatment, estimation
- Trend, seasonal $\rightsquigarrow$ methodology

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Revision policies

ESS guidelines on seasonal adjustment

Goal

- Harmonisation and comparability of the statistics in the European Statistical System (ESS)

Authors

- ECB, Eurostat, ONS, Bundesbank and INSEE

Content

- Theoretical aspects and practical implementation issues
- Alternative options highlighted:
(A) best (B) acceptable (C) to be avoided
- Chapter 6 $\rightsquigarrow$ Revision policies

Revision policies

Current versus concurrent adjustment I/II

Concurrent adjustment (C)

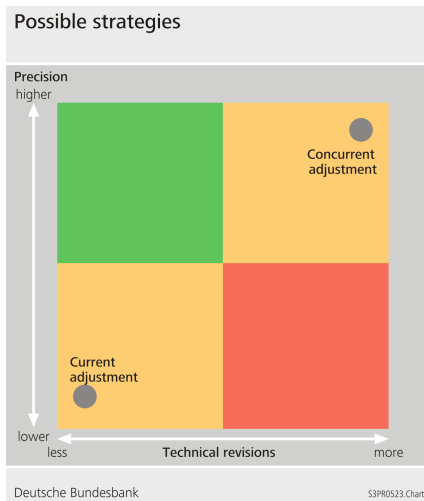
Re-identification of model, filters and outliers and re-estimation of parameters and components every time new or revised data become available

Current adjustment (B)

Re-identification and re-estimation once a year (revision period) and usage of forecasted components in the meantime

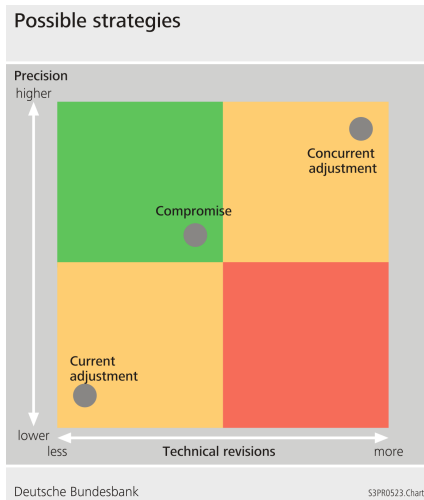
Revision policies

Current versus concurrent adjustment II/II



Revision policies

Recommended approaches I/II



Partial concurrent adjustment (A)

Re-identification once a year, partial re-estimation every time new or revised data become available

Controlled current adjustment (A)

Current adjustment combined with an internal check against the results of the partial concurrent adjustment

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Option	Meaning
Current adjustment (AO approach)	Fix model (including all parameters) handle new observation as AO
Fixed model	Re-estimation, with all parameters fixed
Estimate regression coefficients	Re-estimation of the regression coefficients
+ Arima parameters	and Re-estimation of the parameters of the ARIMA model
+ Last outliers	and Re-estimation of the outliers of the last year only
+ All outliers	and all the outliers
+ Arima model	and Re-estimation of the ARIMA model.
Concurrent	The reference specification is used

Extension of JDemetra+

Basic idea

- Decision whether current factors are still appropriate
- Check current results against partial concurrent

Graphical tools

- Seasonally adjusted series, seasonal components, S-I-ratios

Pre-defined statistics

- Final filters, MSR and I/C-ratio,
- Seasonality tests and heteroscedasticity test
- D8B with current and new D10

HTML-Report

- Summary of figures, tables and statistics (including pre-processing)

How to use CompRes?

Install plug-ins:

- ConCur-1.4.4.nbm
- ConCurReport-1.4.4.nbm

Save to workspace

- Calendar Factor
- Seasonal Factor

Create HTML

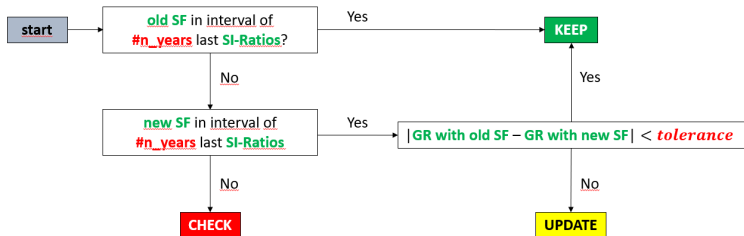
- Summary Report for CompRes

CompRes Recommendations

The Recommendation functionality by the CompRes plugin helps with the decision of whether to keep or not to keep using the previously forecast seasonal component.

Recommendation	Meaning
Check	Check the specification of the SA-Item.
Update	Switch to the new components.
Keep	Continue with the current seasonal and calendar components.

Simplified decision tree of CompRes?

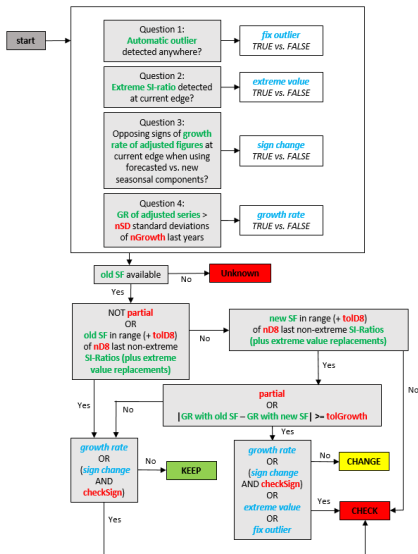


Variables:

- new SF: most recent value in D10 table
- old SF: most recent value in D10 table estimated previously
- SI-Ratios: D8B without extreme values
- GR with new SF: most recent growth rate of D11
- GR with old SF: most recent growth rate of D11 when using old SF instead of new SF

CompRes Recommendations

Decision tree



CompRes Recommendations

Decision tree parameters

- **partial**: Boolean indicating whether the item is adjusted using Partial Concurrent Adjustment or not. Default = **FALSE** (i.e., Controlled Current Adjustment).
- **nSD**: Number of standard deviations used for detecting significant movements at the current edge. Default = 2.0.
- **nGrowth**: Number of years to be considered when calculating standard deviations. Default = 5.
- **nD8**: Number of years used for checking seasonal factors (Note: In the existing **compRes** version, **nD8** is automatically extended by the number of extreme SI ratios at the back). Default = 3.
- **tolD8**: Tolerance within which a seasonal factor may lie outside the interval of the latest SI ratios. If **tolD8** = 0, this corresponds to our previous algorithm. Default = 0.05.
- **tolGrowth**: Acceptable difference in the percentage WR of the adjusted values with previously estimated vs. newly estimated factors. Default = 1.0.
- **checkSign**: Boolean indicating whether sign changes should be considered. Default = **FALSE**.
- **manual**: List of items that should always be checked. Recommendations for these cases are moved to the top and highlighted in black. Default is an empty list.
- **checkPrevious**: Boolean indicating whether the algorithm should also run for the previous period in addition to the current period. In this case, the “stricter” result is returned (**Check** > **Update** > **Keep**). Default = **TRUE**.

CompRes with Excel

Additional plug-in

- ExcelAdapter-1.2.nbm

DataSourceUpdateOption

- `de.bundesbank.jdemetra.exceladapter.excelConnection`

Save to Excel

- Calendar factor
- Seasonal factor
- Forecast

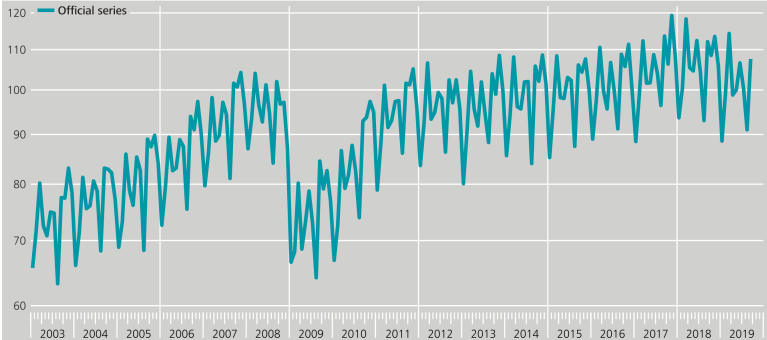
Advantage: use the results without JD+

New SA-View

Example: CompRes

Industrial output of capital goods

2015 = 100, calendar adjusted, log scale



Deutsche Bundesbank

S3PR0498.Chart

New SA-View

Example: CompRes

	jan	feb	mar	apr	may	D88		aug	sep	oct	nov	dec
	jan	feb	mar	apr	may	jun	jul	aug	sep	oct	nov	dec
2009									* 113,573	105,469	109,137	100,451
2010	86,346	* 92,54	108,675	97,921	99,813	105,784	98,184	86,778	107,191	106,071	108,23	* 104,104
2011	85,789	95,131	108,619	97,522	98,194	* 101,772	101,192	88,959	* 104,834	104,485	108,656	99,427
2012	86,548	95,399	110,474	96,638	98,082	102,845	101,561	89,774	107,167	102,231	108,497	101,502
2013	* 84,916	95,825	110,012	* 100,158	* 95,885	105,929	97,938	90,809	106,205	* 100,651	109,865	100,708
2014	86,397	95,388	109,547	97,81	97,675	104,365	* 104,483	* 85,828	107,602	103,306	109,433	101,89
2015	85,525	95,725	108,703	98,394	98,114	103,069	* 102,471	87,836	106,644	104,785	107,779	99,745
2016	* 88,368	96,451	109,216	98,476	* 94,321	105,188	98,23	89,672	106,995	104,047	109,752	* 98,085
2017	86,92	96,934	109,211	98,138	97,554	103,537	98,437	90,471	105,945	* 98,706	* 110,503	100,164
2018	86,671	* 93,086	109,675	97,954	97,56	105,382	98,284	88,224	106,887	103,822	108,889	101,757
2019	85,482	96,279	* 111,156	* 96,41	97,827	104,441	98,516	88,972				
	Seasonal Factor											
	jan	feb	mar	apr	may	jun	jul	aug	sep	oct	nov	dec
2018									106,767	103,827	109,058	100,697
2019	86,371	96,175	109,408	98,045	97,583	104,398	98,699	88,948				
Seasonal Factor (current)												

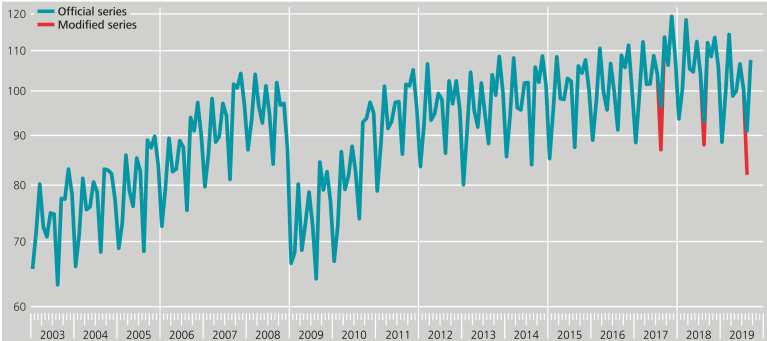
No Data specified for seasonalfactor via Webservice

New SA-View

Example: CompRes

Industrial output of capital goods

2015 = 100, calendar adjusted, log scale



Deutsche Bundesbank

S3PRO498A.Chart

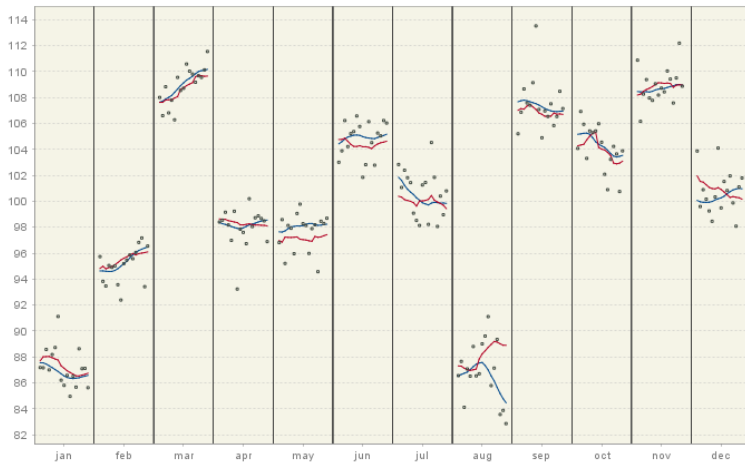
New SA-View

Example: CompRes

	jan	feb	mar	apr	may	D88 jun	jul	aug	sep	oct	nov	dec
2009									* 113,514	105,419	109,056	100,333
2010	86,208	* 92,39	108,536	97,848	99,779	105,759	98,139	86,703	107,074	105,977	108,178	* 104,099
2011	85,813	95,193	108,705	97,604	98,272	* 101,844	101,257	89,013	* 104,896	104,546	108,718	99,482
2012	86,592	95,463	110,575	* 96,735	98,134	102,817	101,445	89,613	106,958	102,077	108,426	101,52
2013	* 84,964	95,861	110,026	* 100,193	* 95,983	106,139	98,218	* 91,108	106,535	* 100,893	110,041	100,826
2014	86,512	95,568	109,807	98,054	97,891	104,515	* 104,531	85,794	107,512	103,228	109,412	101,957
2015	85,678	96,053	109,171	98,73	98,189	102,78	101,847	87,139	105,832	104,225	107,573	99,871
2016	* 88,636	96,83	109,673	98,872	* 94,575	105,251	98,061	* 89,352	106,527	103,646	109,504	* 98,078
2017	87,097	97,17	109,55	98,67	98,458	105,045	100,401	83,572	108,474	* 100,748	* 112,179	101,102
2018	87,12	* 93,418	110,126	98,487	98,281	106,222	98,963	83,881	107,142	103,881	108,871	101,794
2019	85,635	96,551	111,541	* 96,894	98,693	106,027	100,799	82,864				
	jan	feb	mar	apr	may	Seasonal Factor jun	jul	aug	sep	oct	nov	dec
2018									106,942	103,536	108,95	100,961
2019	86,585	96,477	110,165	98,548	98,233	105,167	99,835	84,47				
	jan	feb	mar	apr	may	Seasonal Factor (current) jun	jul	aug	sep	oct	nov	dec
2018									106,696	103,088	108,963	100,164
2019	86,751	96,099	109,649	98,107	97,426	104,627	99,432	88,9				

New SA-View

Example: CompRes



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Summary

Are the current seasonal factors still appropriate?

- Compare current (forecasted) with concurrent (newly estimated) seasonal factors
- Changes in the seasonal pattern?
- Check for outliers and extreme values

Decide on action

- Keep forecasted factors
- Re-estimated factors
- Change specification and re-estimate

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Guidelines



ESS Guidelines



METHODOLOGICAL NOTE GUIDANCE ON TREATMENT OF COVID-19-CRISIS
EFFECTS ON DATA

Plug-ins



<https://github.com/bbkrd/CompRes>



<https://github.com/bbkrd/ExcelAdapter>

Documentation



<https://bbkrd.github.io/>

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